

Week 5: Sinusoidal Sources

Sinusoids, phase, lead and lag, average and RMS values, complex numbers for AC

In-depth notes covering all core material of this chapter: theory, engineering shortcuts, exam traps, 8 worked examples, and 8 practice problems with full solutions.

Based on: Rizzoni & Kearns, *Principles and Applications of Electrical Engineering*, 7th ed., §3.3, plus the complex-number review needed for phasors

Inspired by: ELEC 275, Principles of Electrical Engineering, Concordia University

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How to use this deck

- **“Try it” slides:** the problem alone, with empty space. Do it first.
- **Next slide:** the clean solution, step by step.
- **Purple boxes:** engineering shortcuts, the fast way to see or solve things.
- **Orange boxes:** traps. This is where marks get lost on exams.
- Formulas are tagged **[on formula sheet]** or **[NOT on formula sheet]**, based on the sheet given in ELEC 275 exams.
- **End of deck:** final-check problems, then a full answer sheet.

The right side of each slide is left empty for your notes.

Why sinusoids?

AC is everywhere

The power grid, motors, transformers, radios and audio all use sinusoidal (AC) signals. North America: 60 Hz. Europe: 50 Hz.

Why they're special for circuits

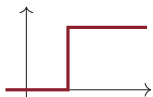
A sinusoid's derivative and integral are also sinusoids at the **same frequency**. So in a circuit with R, L and C, every voltage and current is a sinusoid at the source frequency. Only the **size** and **phase** change.

That second fact is what makes phasors (week 6) work: we only need to track size and phase, which is exactly what a complex number stores.

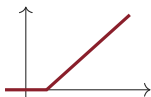
Symbols for this week

Symbol	Unit	Meaning
V_m	V	amplitude (peak value)
f	Hz	frequency: cycles per second
T	s	period: time for one cycle, $T = 1/f$
ω	rad/s	angular frequency, $\omega = 2\pi f$
ϕ	deg or rad	phase angle: where the wave is at $t = 0$
V_{avg} , V_{rms}	V	average value, root-mean-square (effective) value
j		$\sqrt{-1}$ (engineers use j because i is current)

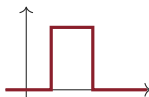
Other time-dependent waveforms



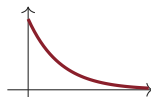
step: switch turns on



ramp



pulse

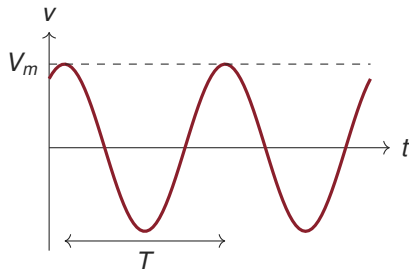


decaying
exponential

These show up in transients (week 4). A pulse is just a step up followed by a step down. The rest of this week is about the sinusoid, the most important waveform in the course.

Anatomy of a sinusoid

[NOT on formula sheet: memorize]



$$v(t) = V_m \cos(\omega t + \phi)$$

$$\omega = 2\pi f \quad \text{rad/s}$$

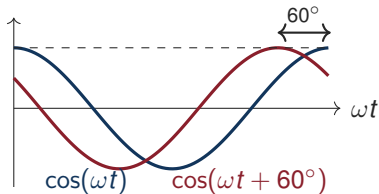
$$f = 1/T \quad \text{Hz}$$

$$\phi \quad \text{phase (shift)}$$

Degrees vs radians

ωt is in **radians**, but ϕ is usually written in **degrees**. Before plugging in a time, convert: $\phi_{rad} = \phi_{deg} \times \pi/180$. Set your calculator's mode to match.

What the phase angle means



ϕ says **where in its cycle** the wave is at $t = 0$.

$\cos(\omega t + 60^\circ)$ reaches every peak 60° **earlier** than $\cos(\omega t)$: its curve is the same shape shifted **left**. So a positive phase means “ahead” (leading).

A negative phase shifts the wave right: later, lagging.

Why sin lags cos by 90° : cos peaks at $\omega t = 0$, sin peaks at $\omega t = 90^\circ$. So $\sin(\omega t) = \cos(\omega t - 90^\circ)$: the same wave, 90° later.

Try it: Example 1, read a sinusoid

$$v(t) = 170 \cos(377t + 30^\circ) \text{ V}$$

Find: (a) amplitude (b) ω (c) f (d) period T (e) phase (f) RMS value

your work here ↓

Example 1: solution

(a) $V_m = 170 \text{ V}$

(b) $\omega = 377 \text{ rad/s}$

(c) $f = \frac{\omega}{2\pi} = \frac{377}{6.283} = 60 \text{ Hz}$

(d) $T = 1/f = 16.7 \text{ ms}$

(e) $\phi = +30^\circ$

(f) $V_{rms} = \frac{170}{\sqrt{2}} = 120 \text{ V}$ (a standard wall outlet)

Shortcut: $\omega = 377$ always means 60 Hz, and $\omega = 314$ means 50 Hz. Worth memorizing.

Phase, and converting sin to cos

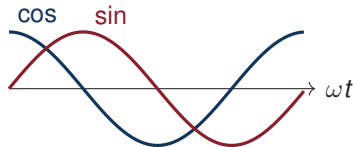
[NOT on formula sheet: memorize]

Put everything in the same form first: positive amplitude and cos

$$\sin(x) = \cos(x - 90^\circ) \quad \text{sin lags cos by } 90^\circ$$

$$-\sin(x) = \cos(x + 90^\circ)$$

$$-\cos(x) = \cos(x \pm 180^\circ) \quad \text{a minus sign is a } 180^\circ \text{ shift}$$



The cos peaks first, and the sin peaks 90° later. So the sin **lags**.

Lead and lag

Comparing two sinusoids at the same frequency

Write both as $A \cos(\omega t + \phi)$ with $A > 0$. The one with the **bigger** phase **leads** (it peaks first) by the difference.

Example: $\cos(\omega t + 50^\circ)$ leads $\cos(\omega t + 20^\circ)$ by 30° .

Phase to time

A phase difference $\Delta\phi$ is a time shift of $\Delta t = (\Delta\phi/360^\circ) T$.

Trap

Phase differences are usually quoted between -180° and $+180^\circ$. “Leads by 270° ” is the same as “lags by 90° ”.

Try it: Example 2, lead, lag and time shift

$$v_1 = 10 \cos(\omega t + 20^\circ), \quad v_2 = 8 \sin(\omega t + 60^\circ), \quad v_3 = -5 \cos(\omega t + 10^\circ)$$

- (a) Rewrite v_2 and v_3 as positive cosines.
- (b) Which leads, v_1 or v_2 , and by how much?
- (c) At $f = 1$ kHz, what time shift is a 45° phase difference?

your work here ↓

Example 2: solution

(a) v_2 $\sin(x) = \cos(x - 90^\circ)$: $v_2 = 8 \cos(\omega t - 30^\circ)$

(a) v_3 $-\cos(x) = \cos(x - 180^\circ)$: $v_3 = 5 \cos(\omega t - 170^\circ)$

(b) $+20^\circ > -30^\circ$: v_1 leads v_2 by 50°

(c) $T = 1 \text{ ms}$, $\Delta t = \frac{45}{360} \times 1 \text{ ms} = 0.125 \text{ ms}$

Average and RMS values

[on formula sheet]

$$V_{avg} = \frac{1}{T} \int_0^T v(t) dt \qquad V_{rms} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$$

RMS = the DC voltage that would heat a resistor exactly the same. So power uses RMS: $P = V_{rms}^2 / R$.

Engineering shortcut: memorize these (NOT on sheet)

Sinusoid	$V_{avg} = 0$	$V_{rms} = V_m / \sqrt{2} \approx 0.707 V_m$
Sawtooth $0 \rightarrow V_m$	$V_m / 2$	$V_m / \sqrt{3}$
Pulse, height V_m , duty D	$D V_m$	$\sqrt{D} V_m$

Where $V_m/\sqrt{2}$ and $V_m/\sqrt{3}$ come from

Sinusoid: $v^2 = V_m^2 \cos^2(\omega t) = \frac{V_m^2}{2} (1 + \cos 2\omega t)$. Over a full period the $\cos 2\omega t$ part averages to zero, so the average of v^2 is $V_m^2/2$:

$$V_{rms} = \sqrt{V_m^2/2} = \frac{V_m}{\sqrt{2}} \approx 0.707 V_m$$

Sawtooth from 0 to V_m in time T : $v = V_m t/T$, so

$$V_{rms}^2 = \frac{1}{T} \int_0^T \frac{V_m^2 t^2}{T^2} dt = \frac{V_m^2}{T^3} \cdot \frac{T^3}{3} = \frac{V_m^2}{3} \Rightarrow V_{rms} = \frac{V_m}{\sqrt{3}}$$

The average of a sinusoid is 0 because the positive and negative halves cancel. That's why AC voltages are described by RMS, not average.

RMS without integrals

Engineering shortcut: RMS of piecewise-constant waves

Square each level, average the squares by time, then take the square root.

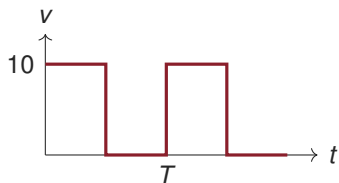
$$V_{rms} = \sqrt{\frac{\sum(\text{level})^2 \times (\text{time at that level})}{T}}$$

Example: +3 V for half the period, −3 V for the other half. The average is 0, but the RMS is $\sqrt{9} = 3$ V.

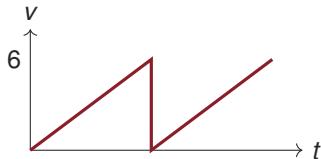
Trap

RMS is **not** the average of the absolute value, and not the average of the peaks. Square first, average, then root, in that order.

Try it: Example 3, average and RMS



(a) square wave, 0 and 10 V, 50% duty



(b) sawtooth, 0 to 6 V

Find the average and RMS value of each waveform.

(c) A heater is rated for 120 V RMS and has $R = 14.4 \, \Omega$. Power? Peak voltage?

your work here ↓

Example 3: solution (1/2)

- (a) avg** 10 V for half the time, 0 V for the other half: $10 \times \frac{1}{2} = 5 \text{ V}$
- (a) rms** square the levels (100 and 0), average by time
($100 \times \frac{1}{2} = 50$), square root: $\sqrt{50} = 7.07 \text{ V}$
- (b) avg** a straight ramp from 0 to 6 spends equal time at every value:
average is the midpoint, $6/2 = 3 \text{ V}$

Example 3: solution (2/2)

(b) rms sawtooth result $V_m/\sqrt{3}$ (derived earlier in this section):

$$6/1.732 = \boxed{3.46 \text{ V}}$$

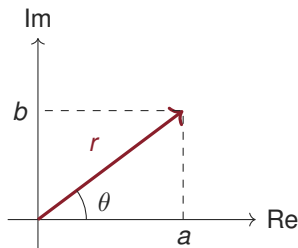
(c) power ratings are RMS, and $P = V_{rms}^2/R = 120^2/14.4 =$

$$\boxed{1000 \text{ W}}$$

(c) peak $V_m = \sqrt{2} V_{rms} = 1.414 \times 120 = \boxed{170 \text{ V}}$

Notice $\text{RMS} \geq \text{average}$ every time: they're only equal for a constant (DC) signal.

Complex numbers: the two forms



Rectangular: $z = a + jb$

Polar: $z = r \angle \theta$

rect $\rightarrow$ polar $r = \sqrt{a^2 + b^2}$, $\theta = \tan^{-1}(b/a)$

polar $\rightarrow$ rect $a = r \cos \theta$, $b = r \sin \theta$

Euler $r \angle \theta = r e^{j\theta}$

The quadrant trap

$\tan^{-1}(b/a)$ only gives angles between -90° and $+90^\circ$. If $a < 0$ (left half), **add** 180° . Always sketch the point first.

Why complex numbers can stand for sinusoids

Euler's formula: $e^{j\theta} = \cos \theta + j \sin \theta$, so $r\angle\theta = re^{j\theta} = r \cos \theta + jr \sin \theta$.

A sinusoid is the **real part** of a rotating complex number:

$$V_m \cos(\omega t + \phi) = \operatorname{Re}\{V_m e^{j\phi} e^{j\omega t}\}$$

All waves in a circuit share the same $e^{j\omega t}$, so only $V_m e^{j\phi} = V_m \angle \phi$ carries information. That's the phasor (week 6).

Why multiply in polar: $r_1 e^{j\theta_1} \cdot r_2 e^{j\theta_2} = r_1 r_2 e^{j(\theta_1 + \theta_2)}$: sizes multiply, angles add. Division: sizes divide, angles subtract.

Why the conjugate trick divides: $(C + jD)(C - jD) = C^2 + D^2$ is real, so multiplying top and bottom by the conjugate removes j from the bottom.

Try it: Example 4, conversions

Convert:

(a) $3 + j4$ to polar

(b) $-3 + j4$ to polar

(c) $10\angle -30^\circ$ to rectangular

your work here ↓

Example 4: solution

(a) $r = \sqrt{9 + 16} = 5$, $\theta = \tan^{-1}(4/3) = 53.13^\circ$: $5\angle 53.13^\circ$

(b) $r = 5$, but $a < 0$ (2nd quadrant): $\tan^{-1}(4/-3) = -53.13^\circ, +180^\circ$
 $5\angle 126.87^\circ$

(c) $10 \cos(-30^\circ) = 8.66$, $10 \sin(-30^\circ) = -5$: $8.66 - j5$

Check (b) with a sketch: $-3 + j4$ is up and to the left, so the angle must be between 90° and 180° . -53° would point down-right: obviously wrong.

Complex arithmetic: pick the easy form

[on formula sheet]

Add / subtract (rectangular) $(a + jb) + (c + jd) = (a + c) + j(b + d)$

Multiply (polar) $r_1 \angle \theta_1 \times r_2 \angle \theta_2 = r_1 r_2 \angle (\theta_1 + \theta_2)$

Divide (polar) $\frac{r_1 \angle \theta_1}{r_2 \angle \theta_2} = \frac{r_1}{r_2} \angle (\theta_1 - \theta_2)$

Divide (rectangular) $\frac{A + jB}{C + jD} = \frac{(A + jB)(C - jD)}{C^2 + D^2}$ (formula sheet)

Conjugate $(a + jb)^* = a - jb, \quad (r \angle \theta)^* = r \angle -\theta$

Handy: $j^2 = -1, \quad 1/j = -j, \quad j = 1 \angle 90^\circ, \quad -j = 1 \angle -90^\circ, \quad -1 = 1 \angle 180^\circ.$

Try it: Example 5, arithmetic

$$A = 3 + j4, \quad B = 1 - j2$$

Find: (a) $A + B$ (b) $A \times B$ (both ways, as a check) (c) A/B

your work here ↓

Example 5: solution (1/2)

(a) adding: real parts together, imaginary parts together:

$$(3 + 1) + j(4 - 2) = 4 + j2$$

(b) **rect** expand like $(x + y)(z + w)$: $3 - j6 + j4 - j^2 8$, and $j^2 = -1$
turns $-j^2 8$ into $+8$: $11 - j2$

(b) **polar** convert: $A = 5\angle 53.13^\circ$, $B = 2.236\angle -63.43^\circ$; multiply
sizes, add angles: $11.18\angle -10.30^\circ$ ✓

Example 5: solution (2/2)

(c) trick multiply top and bottom by the conjugate of the bottom, $1 + j2$, so the bottom becomes $1^2 + 2^2 = 5$ (real)

(c) top $(3 + j4)(1 + j2) = 3 + j6 + j4 + j^2 8 = -5 + j10$

(c) result $\frac{-5 + j10}{5} = \boxed{-1 + j2}$

Preview: adding sinusoids

The idea behind phasors (week 6)

Two sinusoids at the same frequency add to another sinusoid at that frequency. To find it, turn each into a complex number $V_m \angle \phi$, add them, and read off the result.

- 1 write each as $V_m \cos(\omega t + \phi)$
- 2 turn each into $V_m \angle \phi$
- 3 add in rectangular form
- 4 convert back to polar: $R \angle \Phi$ means $R \cos(\omega t + \Phi)$

Try it: Example 6, add two sinusoids

$$v(t) = 10 \cos(\omega t) + 10 \cos(\omega t + 90^\circ)$$

Write $v(t)$ as a single sinusoid $V_m \cos(\omega t + \phi)$.

your work here ↓

Example 6: solution

Plan same frequency, so each term becomes a complex number (its amplitude at its phase), then add

Term 1 $10 \cos(\omega t) \rightarrow 10 \angle 0^\circ = 10 + j0$

Term 2 $10 \cos(\omega t + 90^\circ) \rightarrow 10 \angle 90^\circ = 10 \cos 90^\circ + j10 \sin 90^\circ = 0 + j10$

Add add in rectangular form: $10 + j10$

Polar $r = \sqrt{10^2 + 10^2} = 14.14$, $\theta = \tan^{-1}(10/10) = 45^\circ$ (1st quadrant)

Result back to time: $v(t) = 14.14 \cos(\omega t + 45^\circ)$

Trap: amplitudes don't simply add ($10 + 10 \neq 14.14$). They only add directly when the phases are equal.

Try it: Example 7, sin + cos

$$v(t) = 6 \cos(\omega t) + 8 \sin(\omega t)$$

Write it as a single cosine.

your work here ↓

Example 7: solution

Convert $8 \sin(\omega t) = 8 \cos(\omega t - 90^\circ)$

Complex $6\angle 0^\circ + 8\angle -90^\circ = 6 - j8$

Polar $\sqrt{36 + 64} = 10$, $\tan^{-1}(-8/6) = -53.13^\circ$ (4th quadrant,
 $a > 0$: no fix)

Result

$$v(t) = 10 \cos(\omega t - 53.13^\circ)$$

Shortcut: $A \cos x + B \sin x = \sqrt{A^2 + B^2} \cos(x - \tan^{-1}(B/A))$ when $A > 0$.

Try it: Example 8, exam warm-up

$$i(t) = 170 \sin(377t + 45^\circ) \text{ A}$$

- (a) f and T (b) RMS value (c) Rewrite as a cosine (d) Write it as a complex number $I_m \angle \phi$ (using the cosine form)

your work here ↓

Example 8: solution

(a) $f = 377/2\pi = 60 \text{ Hz}$, $T = 16.7 \text{ ms}$

(b) $170/\sqrt{2} = 120.2 \text{ A}$

(c) $\sin(x) = \cos(x - 90^\circ)$: $170 \cos(377t - 45^\circ)$

(d) $170 \angle -45^\circ$

Step (d) is exactly the phasor you'll use in week 6. Always convert to **cosine** first: the phasor convention in this course is cosine-based.

Traps

Check these before you box an answer

- Degrees vs radians: ωt is radians. Check the calculator mode.
- Compare phases only after both are positive cosines.
- A minus sign in front is a 180° shift, not a negative amplitude.
- $\tan^{-1}$ quadrant: $a < 0$ means add 180° .
- ω (rad/s) vs f (Hz): $\omega = 2\pi f$. Mixing them is the most common slip.
- RMS of a sinusoid is $V_m/\sqrt{2}$. Of other waveforms it is **not**.

Exam-format traps

Is the value peak or RMS?

AC values in problems are sometimes peak ($120 \cos 377t$) and sometimes RMS (“120 V RMS”). Read carefully, and say which one you’re giving.

Units on the angle

Always write $^\circ$ or rad on phase answers. “ -53.13 ” alone is ambiguous.

Show the conversion

When converting sin to cos, write the identity you used. It’s quick and shows the method.

Summary (1/2)

Sinusoid	$V_m \cos(\omega t + \phi)$, $\omega = 2\pi f$, $f = 1/T$; 377 rad/s = 60 Hz
sin / cos	$\sin x = \cos(x - 90^\circ)$, $-\cos x = \cos(x \pm 180^\circ)$
Lead / lag	bigger phase leads; $\Delta t = (\Delta\phi/360^\circ) T$
Avg and RMS	integrals on sheet; sinusoid: avg 0, rms $V_m/\sqrt{2}$; pulse: DV_m , $\sqrt{D}V_m$
Power	$P = V_{rms}^2/R$

Summary (2/2)

Complex forms	$a + jb = r\angle\theta$; $r = \sqrt{a^2 + b^2}$, $\theta = \tan^{-1}(b/a)$ (+180° if $a < 0$)
Operations	add in rectangular; multiply and divide in polar (or conjugate trick)
Adding sinusoids	to complex, add, back to $R\cos(\omega t + \Phi)$
Next week	phasors: this, applied to whole circuits

Final check

8 problems. One per slide, with space. No looking back.

If one takes more than 10 min, go back to that section. The answer sheet comes after the last problem.

P1: read a sinusoid

$$v(t) = 325 \sin(314t - 20^\circ) \text{ V}$$

(a) V_m , f , T (b) V_{rms} (c) the cosine form

your work here ↓

P2: lead and lag

$$i_1 = 4 \cos(\omega t - 10^\circ), \quad i_2 = -3 \sin(\omega t + 40^\circ)$$

(a) Rewrite i_2 as a positive cosine. (b) Which leads, and by how much?

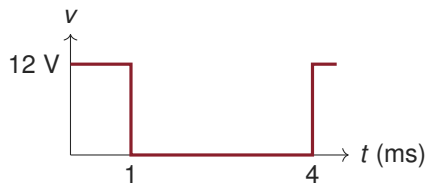
your work here ↓

P3: phase to time

Two 60 Hz voltages are 90° apart. How far apart in time are their peaks?

your work here ↓

P4: average and RMS



12 V for 1 ms, 0 V for 3 ms, repeating.

Find the average and RMS values.

your work here ↓

P5: RMS and power

A lamp draws 60 W from a 120 V RMS outlet.

(a) Its resistance? (b) RMS current? (c) Peak current?

your work here ↓

P6: conversions

(a) $-6 - j8$ to polar. (b) $20\angle 150^\circ$ to rectangular.

your work here ↓

P7: arithmetic

Compute $\frac{(4 + j3)(2 - j2)}{1 + j}$ in rectangular and polar form.

your work here ↓

P8: combine sinusoids

$$v(t) = 5 \cos(\omega t + 30^\circ) + 5 \cos(\omega t - 30^\circ)$$

Write $v(t)$ as a single cosine.

your work here ↓

Answer sheet

Full worked solutions to P1–P8.

P1: solution

Amplitude the number in front: $V_m = 325 \text{ V}$

Frequency $\omega = 314 \text{ rad/s}$, $f = \omega/2\pi = 314/6.283 \approx 50 \text{ Hz}$

Period $T = 1/f = 1/50 = 20 \text{ ms}$

RMS sinusoid: $V_{rms} = V_m/\sqrt{2} = 325/1.414 = 229.8 \text{ V}$

Cosine $\sin(x) = \cos(x - 90^\circ)$ with $x = 314t - 20^\circ$:

$$325 \cos(314t - 110^\circ) \text{ V}$$

P2: solution

Plan compare only after both are written as **positive** cosines

Convert a minus sign in front of a sine: $-\sin(x) = \cos(x + 90^\circ)$

i_2 with $x = \omega t + 40^\circ$: $i_2 = 3 \cos(\omega t + 130^\circ)$

Compare phases $+130^\circ$ (i_2) and -10° (i_1); the bigger phase leads

Result difference $130 - (-10) = 140^\circ$: i_2 leads i_1 by 140°

P3: solution

Idea a full cycle (360°) takes one period T , so a fraction of 360° is the same fraction of T

Period $T = 1/f = 1/60 = 16.67 \text{ ms}$

Fraction $90^\circ/360^\circ = 1/4$

Result $\Delta t = \frac{1}{4} \times 16.67 = 4.17 \text{ ms}$

P4: solution

Read it one period is 4 ms: 12 V for 1 ms, then 0 V for 3 ms, so the duty cycle is $D = 1/4$

Average time-weighted value: $\frac{12 \times 1 + 0 \times 3}{4} = \boxed{3 \text{ V}}$ ($= D V_m$)

RMS step 1 square each level: $12^2 = 144$ and $0^2 = 0$

RMS step 2 average the squares over the period:
$$\frac{144 \times 1 + 0 \times 3}{4} = 36$$

RMS step 3 square root: $\sqrt{36} = \boxed{6 \text{ V}}$ ($= \sqrt{D} V_m = \frac{1}{2} \times 12 \checkmark$)

P5: solution

Key fact ratings like “120 V” and “60 W” are RMS values, and

$$P = V_{rms}^2 / R \text{ for a resistor}$$

$$(a) \quad R = \frac{V_{rms}^2}{P} = \frac{120^2}{60} = \frac{14\,400}{60} = 240 \, \Omega$$

$$(b) \quad I_{rms} = P / V_{rms} = 60 / 120 = 0.5 \, A \quad (\text{check: } 120 / 240 = 0.5 \checkmark)$$

$$(c) \quad \text{peak of a sinusoid} = \sqrt{2} \times \text{RMS: } 0.5 \times 1.414 = 0.707 \, A$$

P6: solution

(a) sketch $-6 - j8$: left and down, so it's in the **3rd quadrant**

(a) size $r = \sqrt{(-6)^2 + (-8)^2} = \sqrt{100} = 10$

(a) angle reference angle $\tan^{-1}(8/6) = 53.13^\circ$; in the 3rd quadrant,
 $\theta = -180^\circ + 53.13^\circ$

(a) result $10\angle -126.87^\circ$ (same as $10\angle 233.13^\circ$; a calculator's $\tan^{-1}(-8/-6)$ would wrongly give 53.13°)

(b) $a = 20 \cos 150^\circ = 20(-0.866) = -17.32,$

$b = 20 \sin 150^\circ = 20(0.5) = 10:$ $-17.32 + j10$

P7: solution

Plan multiply the top out in rectangular form, then divide using the conjugate of the bottom

Top $(4 + j3)(2 - j2) = 8 - j8 + j6 - j^2 6 = 8 - j2 + 6 = 14 - j2$

Conjugate the bottom is $1 + j$, so multiply top and bottom by $1 - j$; the bottom becomes $1^2 + 1^2 = 2$

New top $(14 - j2)(1 - j) = 14 - j14 - j2 + j^2 2 = 12 - j16$

Divide $\frac{12 - j16}{2} = 6 - j8$

Polar $r = \sqrt{36 + 64} = 10$, 4th quadrant ($a > 0$, $b < 0$):
 $\tan^{-1}(-8/6) = -53.13^\circ$: $10 \angle -53.13^\circ$

P8: solution

Plan same frequency, so turn each term into a complex number, add, and turn the sum back into a cosine

Term 1 $5\angle 30^\circ = 5 \cos 30^\circ + j5 \sin 30^\circ = 4.33 + j2.5$

Term 2 $5\angle -30^\circ = 4.33 - j2.5$

Add $(4.33 + 4.33) + j(2.5 - 2.5) = 8.66 + j0 = 8.66\angle 0^\circ$

Result $v(t) = 8.66 \cos(\omega t) \text{ V}$

Why the imaginary parts cancel: two equal waves at $\pm\phi$ always add to a wave at 0°